

Online Examination Report

Date:	13.08.2014
Attempts:	1
Examinee:	Echel, Isaac
Date of birth:	8/7/1991
Subject:	040-Human performance
Result:	97,5 %
Time used:	00:27:14 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	206	40.	HUMAN PERFORMANCE	1,00	1,00
2	170	40.	HUMAN PERFORMANCE	1,00	1,00
3	178	40.	HUMAN PERFORMANCE	1,00	1,00
4	194	40.	HUMAN PERFORMANCE	1,00	1,00
5	184	40.	HUMAN PERFORMANCE	1,00	1,00
6	193	40.	HUMAN PERFORMANCE	1,00	1,00
7	207	40.	HUMAN PERFORMANCE	1,00	1,00
8	169	40.	HUMAN PERFORMANCE	1,00	1,00
9	199	40.	HUMAN PERFORMANCE	1,00	1,00
10	183	40.	HUMAN PERFORMANCE	1,00	1,00
11	203	40.	HUMAN PERFORMANCE	1,00	1,00
12	208	40.	HUMAN PERFORMANCE	1,00	1,00
13	181	40.	HUMAN PERFORMANCE	1,00	1,00
14	20	40.1.2	Accident statistics	1,00	1,00
15	108	40.2	Basic aviation physiology and health maintenance	1,00	1,00
16	63	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
17	67	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
18	49	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
19	100	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
20	18	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
21	52	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
22	102	40.2.2	Man and Environment: the sensory system	1,00	1,00
23	76	40.2.2.1	Central, peripheral and autonomic nervous system	1,00	1,00
24	86	40.2.2.2	Vision	1,00	1,00
25	98	40.2.2.2	Vision	1,00	1,00
26	35	40.2.3.2	Body rythm and sleep	1,00	1,00
27	93	40.2.3.2	Body rythm and sleep	1,00	1,00
28	43	40.2.3.3	Problem areas for pilots	1,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	66	40.2.3.3	Problem areas for pilots	1,00	1,00
30	4	40.2.3.3	Problem areas for pilots	1,00	1,00
31	2	40.3	BASIC AVIATION PSYCHOLOGY	1,00	1,00
32	101	40.3	BASIC AVIATION PSYCHOLOGY	1,00	1,00
33	124	40.3	BASIC AVIATION PSYCHOLOGY	1,00	1,00
34	152	40.3	BASIC AVIATION PSYCHOLOGY	1,00	1,00
35	80	40.3.1.3	Memory	1,00	1,00
36	69	40.3.1.3	Memory	1,00	1,00
37	109	40.3.3	Decision making	1,00	1,00
38	81	40.3.3.1	Decision-making concepts	1,00	1,00
39	128	40.3.5	Human behaviour	1,00	1,00
40	44	40.3.6.2	Stress	0,00	1,00
Total:98%				39,00	40,00

Attachment 2: List of result by topic

Licence

Date: 13.08.2014

Examinee: Echel, Isaac

Location: Uganda Civil Aviation Authority

Your Result: 39,00 from 40,00 Points (97,50 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
40. HUMAN PERFORMANCE	40	39,00	40,00	97,50
40.1. Human Factors: basic concepts	1	1,00	1,00	100,00
40.2. Basic aviation physiology and health maintenance	16	16,00	16,00	100,00
40.3. BASIC AVIATION PSYCHOLOGY	10	9,00	10,00	90,00
Total	40	39,00	40,00	97,50

1

What action should be taken if hyperventilation is suspected?

- ☐ Breathe at a slower rate by taking very deep breaths.
- ☐ Consciously force yourself to take deep breaths and breathe at a faster rate than normal.
- ☒ Consciously breathe at a slower rate than normal.

2

When should a flight instructor begin teaching aeronautical decision making (ADM) to a student?

- ☐ Beginning with the first lesson.
- ☐ After the student has completed the initial solo flight but before conducting cross country flights.
- ☒ As soon as the student is able to control the aircraft during basic maneuvers.

3

Name some hazardous attitudes that can affect your judgment during the aeronautical decision making (ADM) process.

- ☒ **Antiauthority, impulsivity, and resignation.**
- ☐ **Impulsivity, antiestablishment, and reevaluation.**
- ☐ **Peer pressure and stress levels.**

Candidate: Isaac Echel
Date of birth: 1991-08-07

Subject: 040-Human performance
Date: 2014-08-13



4

What visual illusion creates the same effect as a narrower-than-usual runway?

- ☒ **An upsloping runway.**
- ☐ **A downsloping runway.**
- ☐ **A wider-than-usual runway.**

5

What is the antidote for a pilot with a 'macho' attitude?

- ☐ Follow the rules. They are usually right.
- ☐ I'm not helpless. I can make a difference.
- ☒ Taking chances is foolish.

6

Due to visual illusion, when landing on a narrower-than-usual runway, the aircraft will appear to be

- ☐ lower than actual, leading to a higher-than-normal approach.
- ☐ higher than actual, leading to a higher-than-normal approach.
- ☒ higher than actual, leading to a lower-than-normal approach.

7

Which use of cockpit lighting is correct for night flight?

- ☐ The use of regular white light, such as a flashlight, will not impair night adaptation.
- ☒ Reducing the interior lighting intensity to a minimum level.
- ☐ Coloration shown on maps is least affected by the use of direct red lighting.

8

Motion sickness is caused by

- ☐ the movement of an aircraft causing the stomach to create an acid substance which causes the stomach lining to contract.
- ☐ an instability in the brain cells which affects balance and will generally be overcome with experience.
- ☒ continued stimulation of the tiny portion of the inner ear which controls sense of balance.

9

Why is hypoxia particularly dangerous during flights with one pilot?

- ☐ Night vision may be so impaired that the pilot cannot see other aircraft.
- ☐ The pilot may not be able to control the aircraft even if using oxygen.
- ☒ Symptoms of hypoxia may be difficult to recognize before the pilot's reactions are affected.

10

**During a stall recovery, the instructor allows the student to exceed maneuvering speed.
Which best illustrates an 'antiauthority' reaction by the instructor?**

- ☐ The student should know how to recover from a stall by this time.
- ☐ There hasn't been a problem doing this in the past.
- ☒ The aircraft can handle a lot more than the maneuvering speed.

11

How can an instrument pilot best overcome spatial disorientation?

- ☒ Properly interpret the flight instruments and act accordingly.
- ☐ Avoid banking in excess of 30 degrees.
- ☐ Use a very rapid cross-check.

12

Which technique should a pilot use to scan for traffic to the right and left during straight-and-level flight?

- ☐ Concentrate on relative movement detected in the peripheral vision area.
- ☐ Continuous sweeping of the windshield from right to left.
- ☒ Systematically focus on different segments of the sky for short intervals.

13

In the aeronautical decision making (ADM) process, what is the first step in neutralizing a hazardous attitude?

- ☐ Making a rational judgement.
- ☐ Recognizing the invulnerability of the situation.
- ☒ Recognizing hazardous thoughts.

14

The distribution of primary causes of accidents in the worldwide jet aircraft commercial fleet shows that human error is involved in:

- ☐ about 90% of cases
- ☐ all cases, one way or another
- ☒ about 70% of cases
- ☐ 70% of cases

15

Hypoxia susceptibility due to inhalation of carbon monoxide increases as

- ☒ altitude increases.
- ☐ oxygen demand increases.
- ☐ humidity decreases.

16

When flying above 10.000 feet hypoxia arises because:

- ☐ composition of the air is different from sea level
- ☐ composition of the blood changes
- ☒ partial oxygen pressure is below the critical value of 55mm/Hg
- ☐ percentage of oxygen is lower than at sea level

17

Flying immediately following a dive with SCUBA diving equipment to a depth greater than 10 metres

- ☐ has no influence on altitude flights
- ☐ presents any dangers caused by DCS (decompression sickness) when climbing to altitudes not exceeding 30 000 FT
- ☒ can cause decompression sickness even when flying at pressure altitudes below 18 000 FT
- ☐ will always lead to hypoxia

18

Which of the following statements are correct:

- 1: SCUBA diving may be practiced without restriction
- 2: Many medicines have effects which are incompatible with flight safety
- 3: An adequate amount of fluid should be drunk when flying
- 4: Alcohol has no effect on the inner ear.

☐ 2, 3 and 4 are correct

☐ 1, 2 and 3 are correct

☒ 2 and 3 are correct

☐ 1, 3 and 4 are correct

Candidate: Isaac Echel
Date of birth: 1991-08-07

Subject: 040-Human performance
Date: 2014-08-13



19

Positive G will cause the blood flow to the brain to:

☐ increase

☐ remain constant

☒ decrease

☐ first increase, then decrease

20

The rate and depth of breathing is primarily controlled by:

- ☐ the amount of carbon monoxide in the blood
- ☐ the total atmospheric pressure
- ☐ the amount of nitrogen in the blood
- ☒ the amount of carbon dioxide in the blood

21

What can be said concerning the following two statements?

1. Euphoria can be a symptom of hypoxia.
2. Someone in an euphoric condition is more prone to error.

- ☐ 1 and 2 are both false
- ☒ 1 and 2 are both correct
- ☐ 1 is false, 2 is correct
- ☐ 1 is correct, 2 is false

22

To scan properly for traffic, a pilot should

- ☒ use a series of short, regularly spaced eye movements that bring successive areas of the sky into the central visual field.
- ☐ concentrate on any peripheral movement detected.
- ☐ slowly sweep the field of vision from one side to the other at intervals.

23

A stereotype and involuntary reaction of the organism on stimulation of receptors is called:

☐ control system

☐ data processing

☒ reflex

☐ change of stimulation level

24

The retina of the eye

- ☐ filters the UV-light
- ☐ only regulates the light that falls into the eye
- ☐ is the muscle, changing the size of the crystalline lens
- ☒ is the light-sensitive inner lining of the eye containing the photoreceptors essential for vision

25

The retina allows for colour perception as a result of the:

- ☐ crystalline lens
- ☐ rods located in its central part
- ☒ cones located in its central part
- ☐ rods located in its peripheral zone

26

The physiological rhythms of a pilot in a new time zone will resynchronise to this new time zone at a rate of about

☐ 4 - 4.5 hours a day

☒ 1 - 1.5 hours a day

☐ 3 - 3.5 hours a day

☐ 2 - 2.5 hours a day

27

which of the following best describes disturbance of the biological clock?

1. bad night's sleep
2. day flight Amsterdam - New York
3. day flight Amsterdam - Johannesburg
4. night flight New York - Amsterdam

☐ 1,2 and 3 are correct

☐ 1,2,3 and 4 are correct

☒ 2 and 4 are correct

☐ 1 and 3 are correct

Candidate: Isaac Echel
Date of birth: 1991-08-07

Subject: 040-Human performance
Date: 2014-08-13



28

Medical conditions such as high blood pressure, coronary problems and diabetes are associated with:

☐ hypoxia

☒ obesity

☐ cholera

☐ anorexia nervosa

Candidate: Isaac Echel
Date of birth: 1991-08-07

Subject: 040-Human performance
Date: 2014-08-13



29

Loss of elasticity of the crystalline lens due to ageing occurs in:

☐ astigmatism

☒ presbyopia

☐ hypermetropia

☐ myopia

30

How is Yellow fever contracted?

- ☐ Contaminated food or water.
- ☐ Excessive use of alcohol.
- ☒ A virus transmitted by an infected mosquito.
- ☐ By contact with the saliva of infected animals.

Candidate: Isaac Echel
Date of birth: 1991-08-07

Subject: 040-Human performance
Date: 2014-08-13

31

What are two types of attention ?

☐ Divided and behavioural

☐ Intuitive and behavioural

☒ Selective and divided

☐ Cognitive and intuitive

32

As hyperventilation progresses, a pilot can experience

- ☐ decreased breathing rate and depth.
- ☒ symptoms of suffocation and drowsiness.
- ☐ heightened awareness and feeling of well being.

33

While conducting an operational check of the cabin pressurization system, the pilot discovers that the rate control feature is inoperative. He knows that he can manually control the cabin pressure, so he elects to disregard the discrepancy. Which of the following alternatives best illustrates the INVULNERABILITY reaction?

- ☐ He can handle a little problem like this.
- ☐ It's too late to fix it now.
- ☒ What is the worst that could happen.

34

To communicate effectively, instructors must

- ☐ display an authoritarian attitude.
- ☐ utilize highly organized notes.
- ☒ display a positive, confident attitude.

35

Working memory enables us, for example,

- ☐ to remember our own name
- ☒ to remember a clearance long enough to write it down
- ☐ to ignore messages for other aircraft
- ☐ to store a large amount of visual information for about 0.5 seconds

36

Which of the following is true with reference to semantic memory?

- ☐ It is the memory of events; it lasts for a longer period of time than episodic memory and tends to be more accurate.
- ☐ It is the memory of events; it lasts for a shorter period of time than episodic memory and tends to be less accurate.
- ☒ It is the meaning of words and general knowledge of the world; it lasts for a longer period of time and tends to be more accurate than episodic memory.
- ☐ It is the meaning of words and general knowledge of the world; it lasts for a shorter period of time than episodic memory and tends to be less accurate.

37

on which features does risk management, as part of the Aeronautical Decision Making (ADM) process, rely?

- ☒ Situational awareness, problem recognition, and good judgment.
- ☐ Application of stress management and risk element procedures.
- ☐ The mental process of analyzing all information in a particular situation and making a timely decision on what action to take.

38

Judgement is based upon:

- ☐ a decision-making process involving the 5 physical senses and their use to manually operate the aircraft controls
- ☐ the ability to interpret the flight instruments
- ☐ the development of skills through constant practice of flight manoeuvres
- ☒ a process involving a pilot's attitude to take and to evaluate risks by assessing the situation and making decisions based upon knowledge, skill and experience

39

A pilot and friends are going to fly to an out-of-town football game. When the passengers arrive, the pilot determines that they will be over the maximum gross weight for takeoff with the existing fuel load. Which of the following alternatives best illustrates the RESIGNATION reaction?

- ☒ Well, nobody told him about the extra weight.
- ☐ He can't wait around to de-fuel, they have to get there on time.
- ☐ Weight and balance is a formality forced on pilots by the FAA.

40

When can the General Adaption Syndrome(GAS) activate?

- ☐ Responding to Hypovigilance.
- ☒ Responding to an imaginary threat.
- ☐ Responding to boredom.
- ☒ Responding to Circadian Disrhythmia.